GALGOTIAS COLLEGE OF ENGINEERING & TECHNOLOGY (GCET)

Department of Computer Applications

MCA Mini Project (2026-27)

PROPOSED MINI PROJECT IDEAS / TITLE SUBMISSION FORM

Submission Deadline: On or Before 21 August 2026

Project Group No.	**B2 – G9**	**Section**	**B**
Project Supervisor	Ms. Jyoti Pasi	**Project Group Leader**	NITISH KUMAR YADAV

Names and Roll numbers of Group Members:

1. Nitish Kumar Yadav **Roll No.**2500970140124 2. Payal Kumari Roll No. 2500970140128 3. Rahematullah Roll No. 2500970140143

Three Proposed Project Ideas (*Attach Separate Sheet If Needed***)**

Details	**Idea 1**	**Idea 2**	**Idea 3**
Proposed Project Title	**ReCallQ** – AI-Powered Digital Memory & Intelligent File Management	**KaushalSetu** – Verifiable Digital Professional Identity & Reputation Platform for Skilled Workers	**KrishiMitra** – AI-Powered Farmer Decision & Agricultural Information Platform
Problem Statement / Need	Users often struggle to find their saved digital files because they forget the file name or location, while duplicate files create unnecessary digital clutter and storage usage.	Skilled workers often lack verifiable digital records of their skills and work experience, making it difficult for employers to assess and trust their capabilities.	Farmers often lack timely information about the fair value of their crops, suitable buyers, and government schemes they may be eligible for.
Key Objectives (2-3)	1. Enable natural-language searching and retrieval of digital files based on their content and context. 2. Automatically understand and organize files using AI, OCR and semantic analysis. 3. Detect duplicate and redundant files to reduce digital clutter and unnecessary storage usage.	1. Create a digital professional profile containing verified skills, certifications and work history. 2. Build a trustworthy reputation system using verified work, assessments and employer or customer feedback. 3. Connect skilled workers with suitable employers and job opportunities based on verified capabilities.	1. Provide farmers with data-driven estimates of crop value and compare available selling options. 2. Help farmers identify suitable markets or buyers by considering price, location and transportation costs. 3. Personalize government-scheme information based on the farmer's profile, eligibility and requirements.
Proposed Solution / Major Features	ReCallQ is an AI-powered intelligent file management system that uses OCR, document understanding, embeddings and semantic search to understand files. Users can search using natural-language queries such as "Find my internship screenshot" or "Show my college notes PDF." The DEDUPEX module detects duplicate and redundant files and helps users manage digital clutter.	KaushalSetu provides a digital Skill Passport containing a worker's skills, certifications, experience and verified work history. Major features include QR-based credential verification, skill assessments, trust/reputation scoring, employer dashboards and intelligent job matching. Voice and regional-language onboarding can improve accessibility for workers with limited digital literacy.	KrishiMitra is an AI-powered agricultural decision-support platform that combines farmer, crop and market information to provide personalized recommendations. It can provide estimated fair-price ranges, compare buyer and mandi options, calculate estimated net returns after transportation costs, and recommend relevant government schemes. A multilingual and voice-based assistant can improve accessibility for farmers.
Target Users / Beneficiaries	Students, professionals, researchers and individuals who manage large collections of digital documents, photos,	Electricians, plumbers, carpenters, mechanics, technicians, welders and other skilled workers, along	Small and marginal farmers, individual farmers, Farmer Producer Organizations (FPOs), agricultural

	screenshots, notes and personal files.	with contractors, small businesses, employers and skill-training institutes.	cooperatives and rural agricultural support organizations.
Expected Outcome	Faster and more accurate file retrieval, improved digital organization, reduced duplicate files and better utilization of digital storage.	Improved employability of skilled workers, faster and more reliable hiring, reduced skill and credential mismatch, better verification of professional experience and creation of a portable digital professional reputation.	Improved awareness of crop value and market opportunities, better selling decisions, reduced information asymmetry, improved access to relevant government schemes and increased accessibility of agricultural information.
Proposed Technologies / Tools	Python, FastAPI, React/JavaScript, OCR, NLP, Artificial Intelligence, Machine Learning, Embeddings, Semantic Search, Vector Database, Document Processing, PostgreSQL and Duplicate Detection.	React/Next.js, Python/FastAPI, PostgreSQL, REST APIs, Secure Authentication, OCR, NLP, Machine Learning, Recommendation Systems, QR-based Verification and Verifiable Digital Credentials.	Python, FastAPI, React/JavaScript, Machine Learning, NLP/LLM, Recommendation Systems, Market Data APIs, Geolocation/Maps, PostgreSQL, Multilingual Voice Processing and Data Analytics.
Relevant UN SDG(s)	SDG 9 – Industry, Innovation and Infrastructure SDG 4 – Quality Education SDG 12 – Responsible Consumption and Production	SDG 8 – Decent Work and Economic Growth SDG 4 – Quality Education SDG 9 – Industry, Innovation and Infrastructure SDG 10 – Reduced Inequalities	SDG 2 – Zero Hunger SDG 8 – Decent Work and Economic Growth SDG 9 – Industry, Innovation and Infrastructure SDG 10 – Reduced Inequalities
SDG Alignment / Justification	Supports AI-driven innovation in digital information management, improves access to educational resources for students, and promotes responsible management of digital resources by identifying duplicate and redundant files.	Promotes decent employment and skill development, improves access to job opportunities, reduces information gaps between workers and employers, and uses digital technology to support inclusive economic participation.	Supports farmer livelihoods and food security by improving access to agricultural information, promotes inclusive digital innovation, and helps reduce information gaps affecting rural communities.
Scope for Major Project Extension	Cloud and mobile synchronization, multimodal search, personal knowledge graphs, smart backup, document summarization, cross-device memory and privacy-preserving or on-device AI processing.	Expansion to multiple professions and cities, multilingual voice-based onboarding, advanced skill assessments, personalized training recommendations, employer APIs, career analytics and integration with verified training and credential providers.	Weather and risk alerts, crop disease assistance, demand forecasting, storage and sell-later recommendations, direct buyer networks, crop insurance assistance, agricultural credit information and advanced farm-management analytics.
Original / Distinct Feature	Combines context-aware semantic file retrieval with the DEDUPEX duplicate detection module, allowing users to find files by meaning and context rather than relying only on filenames, folders or keywords.	KaushalSetu creates a portable professional reputation based not only on certificates but also on verified real-world work history, skills, assessments and performance, allowing workers to carry their professional identity across employers and locations.	KrishiMitra does not simply display mandi prices or government schemes. It converts scattered agricultural information into personalized decision support, helping farmers understand estimated crop value, potential net returns, suitable selling options and relevant government benefits.

Supervisor Recommendation / Departmental Processing

Recommended Preference	1st: Idea ___	2nd: Idea ___	3rd: Idea ___

Supervisor Remarks	
Final Title Approved / Selected	

Signature: _____ Date : _____	Signature: _____ Date : _____	Signature: _____ Date : _____
Member 1	**Member 2**	**Member 3**

Signature: _____ Date : _____	Signature: _____ Date : _____	Signature: _____ Date : _____
Project Supervisor **Name:** Ms. Jyoti Pasi	**Project Coordinator** Dr. Prashant Kumar Nag	**Head of Department (HOD)** Prof. (Dr.) Vinit Kumar

Note: All three ideas must be discussed with the assigned Supervisor. SDG mapping should be relevant and justified. Proposed ideas should be original, feasible, and suitable for possible extension into the Major Project.